BIRZEIT UNIVERSITY

COMP233 – Discrete Mathematics

First Semester 2026/2027 (1261)

Quiz 1 – Form D: Answer Key

Instructor copy – do not distribute to students

Total: 100 marks. Accept mathematically equivalent answers and valid alternative proofs. A bare answer does not earn marks allocated to justification.

Question 1 [20 marks]

(a) Truth table and classification. *[15 marks]*

$$H = (p \to q) \to (\sim p \to \sim q).$$

p	q	$\sim p$	$\sim q$	$p \to q$	$\sim p \to \sim q$	H
T	T	F	F	T	T	T
T	F	F	T	F	T	T
F	T	T	F	T	F	F
F	F	T	T	T	T	T

H is **neither a tautology nor a contradiction** (it is contingent): its final column contains both T and F.

Marking: All four distinct input assignments: 2. Intermediate and final truth values: 8 (2 per row: 1 for the two intermediate implications and 1 for H). Classification: 2. Justification that the final column contains both truth values: 3. Total: 15. Accept any row order. Omission of auxiliary negation columns alone is not an error when the implication columns and H are correct.

(b) Sufficient and necessary conditions. *[5 marks; 2.5 each]*

(i) Sufficient condition: $\boxed{p \to q}$. A complete backup is sufficient for restoration.

(ii) Necessary condition: $\boxed{q \to p}$. Restoration requires a complete backup.

Marking: Award 2.5 for each correct implication or logically equivalent formula. Accept, for example, $\sim q \to \sim p$ for (i) and $\sim p \to \sim q$ for (ii). Mark the two statements independently; a single $p \leftrightarrow q$ is not an answer to either individual statement. Explanations are not required.

Question 2(a) [15 marks]

Simplification by logical equivalences.

$$\begin{aligned}
&(\sim p \to q) \wedge (\sim p \to \sim q) \\
&\quad \equiv (\sim\sim p \vee q) \wedge (\sim\sim p \vee \sim q) && \text{Conditional equivalence} \\
&\quad \equiv (p \vee q) \wedge (p \vee \sim q) && \text{Double negative law} \\
&\quad \equiv p \vee (q \wedge \sim q) && \text{Distributive law} \\
&\quad \equiv p \vee \mathrm{F} && \text{Negation law} \\
&\quad \equiv \boxed{p} && \text{Identity law.}
\end{aligned}$$

Marking: Five essential transformations, 3 each: 2 for the transformation and 1 for its law. Equivalent proofs may combine or reorder steps; award by mathematical content, not the number of written lines. A truth table alone does not meet the requested method.

Question 2(b) [20 marks]
Numbered inference proof.

Line	Statement	Reason
1	$p \vee q$	Premise
2	$q \to r$	Premise
3	$\sim r \wedge s$	Premise
4	$\sim r$	Specialization (conjunction elimination), line 3
5	$\sim q$	Modus tollens, lines 2 and 4
6	$\boxed{p}$	Elimination (disjunctive syllogism), lines 1 and 5

The unused component s does not need a separate derived line.

Marking: Each of lines 4–6: 6 marks (correct statement 2, correct rule 2, correct supporting references 2), total 18. A coherent numbered proof with the premises clearly identified: 2. Total: 20. Accept any valid alternative rule-based proof. A truth table alone does not meet the requested method.

Question 3(a) [10 marks]
Formalization.

$$\boxed{\forall x \in L\,(R(x) \to \sim D(x)).}$$

Equivalent accepted answers include:

$$\sim\exists x \in L\,(R(x) \wedge D(x)), \qquad \forall x \in L \sim (R(x) \wedge D(x)),$$

$$\forall x \in L\,(D(x) \to \sim R(x)).$$

Marking: Appropriate quantification and scope: 4. Correct relationship between the predicates: 6. The question does not prescribe a quantifier; award full credit to any logically equivalent formalization, including a negated existential. Unrestricted quantifier notation is acceptable because the domain is stated. Do not accept $\forall x\,(R(x) \wedge \sim D(x))$, which incorrectly asserts that every item is reserved.

Question 3(b) [10 marks]
Symbolic negation.

$$\begin{aligned} &\sim[\forall x \in L\,(R(x) \to \sim D(x))] \\ &\quad \equiv \exists x \in L \sim (R(x) \to \sim D(x)) \\ &\quad \equiv \exists x \in L\,(R(x) \wedge \sim\sim D(x)) \\ &\quad \equiv \boxed{\exists x \in L\,(R(x) \wedge D(x)).} \end{aligned}$$

Marking: Existential quantification and correct scope: 4. Correct conjunction of the positive predicates: 6. A correct simplified final formula is sufficient; neither a derivation nor an English restatement is required. Accept equivalent simplified formulas.

Linked-part grading: Mark part (a) separately. In part (b), credit correct negation steps on a student's explicitly stated formula from (a), but full credit requires the negation of the printed statement. Do not repeatedly penalize the same notational slip.

Question 3(c) [15 marks]

Interpret and evaluate S.

$$S: \quad \forall u \in U\, \exists f \in F \sim A(u, f).$$

English: "Every user has at least one file that they cannot access."

The statement is **true**. For Ala, choose f_2, since $A(\text{Ala}, f_2) = \text{F}$. For Basma, choose f_1, since $A(\text{Basma}, f_1) = \text{F}$. Thus each user has a file they cannot access; the chosen files need not be the same.

Marking: Translation: 5 (every user 2, at least one file 1, correct negative predicate 2). Truth value and justification: 10 (true 2, a valid file and its justification for Ala 4, a valid file and its justification for Basma 4). Total: 15. "No user can access any file" is not an equivalent translation.

Question 3(d) [10 marks]

Symbolic negation of S.

$$\begin{aligned} \sim S &\equiv \exists u \in U \sim(\exists f \in F \sim A(u, f)) \\ &\equiv \exists u \in U\, \forall f \in F \sim\sim A(u, f) \\ &\equiv \boxed{\exists u \in U\, \forall f \in F\, A(u, f).} \end{aligned}$$

Marking: Existential user quantifier: 3. Universal file quantifier: 3. Positive predicate after double negation: 2. Correct nesting, binding, and domains: 2. A correct final formula is sufficient; neither derivation steps nor an English restatement are required.

Assessment summary

Question	Marks	Assessed skills
1	20	Truth table and classification (15); sufficient and necessary conditions (5).
2	35	Equivalence laws (15); numbered inference proof (20).
3	45	Formalization (10); symbolic negation (10); interpretation and finite-model evaluation (15); nested symbolic negation (10).
Total	**100**	**30 minutes; three questions.**

Planning allocation: Q1: 8 minutes; Q2: 8 minutes; Q3: 11 minutes; reading and checking: 3 minutes. These are planning estimates, not measured student completion times.

Source basis: COMP233 course outline, first semester 2026/2027, p. 2 (Quiz 1 scope); supplied Chapters 2–3 quiz, p. 1 (benchmark); Epp, 4th edition, Sections 2.1–2.3 and 3.1–3.3; local Jarrar notes 2.2–2.3, 3.1–3.3. The coordinator-approved revision and requested extra Q1 branch govern the selected tasks. Layout follows the supplied BZU exam-source template.